CYBERSECURITY THREAT INTELLIGENCE REPORT

Cybersecurity & Ethical Hacking | Task 1

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Awareness and research project: five modern cyber threats, their impacts, real-world examples, and practical defensive measures.

Prepared for learning and public portfolio sharing.

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Each threat section answers the same four assignment prompts: what the threat is, how it affects individuals and organizations, a real-world case, and preventive measures.

1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the practice of protecting computers, mobile devices, networks, software, cloud services, and data from unauthorized access, misuse, disruption, or destruction. It combines technical safeguards—such as access controls, encryption, and monitoring—with safe procedures and informed user behavior. Its core aims include confidentiality (information is seen only by authorized people), integrity (information remains accurate), and availability (systems and data can be used when needed).

1.2 Why is Cybersecurity Important for Individuals and Businesses?

For individuals, cybersecurity helps protect identity, personal photographs, messages, health details, financial accounts, and devices from theft, fraud, surveillance, or disruption. A stolen password can expose multiple accounts when passwords are reused. For businesses, cybersecurity protects customer and employee records, intellectual property, operations, income, and trust. A successful incident can interrupt services, create recovery and legal costs, trigger reporting obligations, and harm customers and suppliers. Cybersecurity is therefore part of reliable service delivery and responsible data handling, not only an IT concern.

1.3 What Makes Cybersecurity Relevant Today?

Daily life and business rely on connected services, remote access, cloud platforms, mobile applications, and Internet of Things (IoT) devices. Each connection can add accounts, software, and settings that need protection. Cybercrime groups also share tools and services, while artificial intelligence can help produce convincing messages and impersonations at scale. Vulnerabilities in widely used software can be exploited before every organization has time to patch. These trends make prevention, timely updates, independent verification, and tested recovery plans essential.

2. Five Major Modern Cyber Threats

2.1 AI-Powered Phishing Attacks

What is the threat?

Phishing uses deceptive messages to persuade people to reveal credentials, open harmful attachments, approve a login, or send money. AI can help criminals write polished, personalized messages, translate them, and imitate a person’s voice or appearance in audio or video deepfakes. The technology makes impersonation easier to scale, but the attack still relies on trust, urgency, and a request that bypasses normal checks.

Impact on individuals

A person may lose money, have accounts taken over, or expose identity and private information. A compromised email or social account can then be used to deceive friends, family, or coworkers.

Impact on organizations

Organizations may face fraudulent payments, stolen credentials, exposed data, account takeover, service disruption, and reputational harm. A single employee’s compromised account can be used to target additional staff or customers.

Real-world case study

In January 2024, Hong Kong police reported a case involving a fabricated video conference with deepfake versions of colleagues. A staff member was instructed to transfer nearly HK$4 million. The case showed how a familiar meeting format can be used to reinforce a false payment request.

Preventive measures

Verify unusual payment or password requests using a separate, previously known contact channel. Require two-person approval and call-back checks for high-value transfers or changes to supplier bank details. Use phishing-resistant multi-factor authentication (MFA), email protections, and practical staff training that teaches people to pause and report suspicious requests.

2.2 Ransomware-as-a-Service (RaaS)

What is the threat?

Ransomware encrypts or blocks access to systems and data. In a RaaS model, developers provide ransomware, infrastructure, or negotiation services to affiliates, who carry out intrusions and share proceeds. Criminals may also steal data before encrypting systems, then threaten to publish it. Initial access can come from stolen credentials, phishing, or unpatched systems.

Impact on individuals

People may lose access to services, have personal or health information exposed, face identity theft, or experience interrupted care when a critical provider is affected.

Impact on organizations

Organizations can suffer prolonged downtime, data loss, recovery expense, extortion pressure, legal and regulatory scrutiny, and damage to public confidence. Disruption to a central service can affect many customers and partner organizations at once.

Real-world case study

Change Healthcare experienced a major ransomware incident in February 2024. UnitedHealth Group testimony said criminals used compromised credentials to access a remote access portal that did not have MFA; ransomware was deployed later. The resulting disruption affected claims and payment processes across healthcare, showing how dependent services can spread impact widely.

Preventive measures

Require MFA on all remote and privileged access. Keep offline or immutable backups and regularly test restoring systems from them. Segment networks, restrict administrator privileges, patch exposed systems promptly, and rehearse an incident plan that includes technical recovery, legal review, communications, and continuity of essential services.

2.3 Cloud Security Misconfigurations

What is the threat?

Cloud misconfiguration occurs when cloud storage, identities, networks, or services are set up with unsafe access or security settings. Examples include publicly accessible storage, excessive permissions, exposed credentials, and incomplete logging. Cloud providers protect the underlying infrastructure, but customers remain responsible for many identity, data, and configuration choices under the shared-responsibility model.

Impact on individuals

Misconfigured services can expose names, contact details, financial or health information, and account credentials. People may then face fraud, targeted phishing, or misuse of personal information.

Impact on organizations

Organizations may experience data theft, service abuse, investigation and notification costs, regulatory action, litigation, and reputational damage. Closing public access does not retrieve copies that may already have been taken.

Real-world case study

In July 2019, Capital One disclosed unauthorized access to personal information belonging to credit-card applicants and customers. Capital One said the attacker exploited a configuration vulnerability and that it fixed the issue. Though earlier than the report period, the incident remains a useful example of how application and cloud access design can expose sensitive records.

Preventive measures

Block public access by default and review every exception. Apply least-privilege permissions and MFA for administrator accounts. Use automated configuration checks, accurate asset inventories, encryption, centralized audit logs, and alerts for risky changes or unusual data access.

2.4 IoT Vulnerabilities

What is the threat?

IoT devices include routers, cameras, video recorders, sensors, and smart appliances. Weak default passwords, unpatched firmware, exposed management interfaces, and short support lifetimes can leave them vulnerable. Attackers may compromise many devices to form a botnet, which can be used for denial-of-service attacks, surveillance, or as a stepping stone into other networks.

Impact on individuals

Compromised home cameras or routers can threaten privacy, expose home network activity, or be used in attacks without the owner's knowledge. Poorly secured smart devices may also provide a route to other personal devices.

Impact on organizations

Organizations can face outages, stolen information, unauthorized network access, and operational or safety risks where connected devices support business or industrial processes. A compromised device fleet can also harm other internet users through botnet activity.

Real-world case study

In September 2024, the FBI announced disruption of a large botnet operated by China-linked Flax Typhoon actors. The network included compromised internet-connected devices such as storage devices, cameras, and video recorders. The incident illustrated how ordinary edge devices can be assembled into infrastructure for broader malicious activity.

Preventive measures

Change default passwords to unique credentials and disable services that are not needed. Install firmware updates and replace devices that no longer receive security support. Isolate IoT devices on a separate network, limit their access to sensitive systems, and monitor unusual outbound connections.

2.5 Zero-Day Exploits

What is the threat?

A zero-day exploit takes advantage of a software vulnerability before the affected organization has a working fix or sufficient time to apply one. Attackers may use the flaw to bypass authentication, run commands, steal data, or establish persistent access. After a flaw becomes public, defenders must identify affected assets, apply vendor mitigations or patches, and check whether an attacker already gained access.

Impact on individuals

Individuals may be affected when a vulnerable service exposes accounts or personal data, or when a compromised provider interrupts services they rely on. Targeted attacks may also lead to identity theft or privacy loss.

Impact on organizations

Organizations risk unauthorized access, espionage, ransomware, data loss, and service interruption. Internet-facing devices are especially concerning because they may connect directly to internal networks and be difficult to monitor.

Real-world case study

In early 2024, CISA and partner agencies reported active exploitation of vulnerabilities in Ivanti Connect Secure and Policy Secure gateways. The flaws could be chained to bypass authentication and execute commands. Agencies issued mitigation and remediation guidance, highlighting the need to respond rapidly to exploited edge-device flaws.

Preventive measures

Maintain an inventory of internet-facing software and subscribe to vendor and CISA alerts. Prioritize vulnerabilities known to be exploited and apply patches or vendor mitigations quickly. Segment networks, monitor endpoints and gateway logs, and investigate devices for signs of compromise rather than assuming a patch alone removes an intruder.

3. Conclusion and Future Scope

Cybersecurity is important because people and organizations depend on digital systems to communicate, work, manage money, and deliver essential services. The five threats in this report show that cyber incidents can begin with a convincing message, a missing access control, an exposed cloud setting, an insecure connected device, or a flaw without an available patch. Their consequences can include financial loss, data exposure, downtime, and reduced trust.

Proactive security reduces the chance that one weakness becomes a lasting crisis. Organizations should keep systems updated, limit access, use MFA, maintain tested backups, monitor for suspicious activity, and practice incident response. Individuals should use unique passwords, enable MFA, update devices, verify unusual requests independently, and report suspected fraud quickly. Because technology and attacker methods continue to change, continuous learning and regular review of defenses are essential.

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6. CISA, Cross-Sector Cybersecurity Performance Goals.

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